

NOVA CONGLOMERATE

Complete Business Analysis & Operational Optimization Capstone

Multi-Divisional Performance Portfolio for Retail, Telecom, and Property Units

Prepared For:	Nova Conglomerate Board of Directors
Prepared By:	Data Science Internship Program - Week 8
Date:	July 1, 2026
Classification:	Confidential - Internal Use Only

1. Project Scope & Business Framework

Nova Conglomerate is a multi-sector enterprise operating across retail, telecommunications, and real estate markets. This capstone report represents a rigorous analytical assessment of three primary operating divisions. By examining transaction-level retail patterns, customer subscription behaviors, and property valuations, this portfolio identifies optimizations that drive top-line growth, combat customer churn, and refine asset valuation models.

Division	Dataset	Business Focus	Primary KPIs
Nova Retail	sales_data.csv (100 rows)	Product/Region Sales, Elasticity	Total Revenue, AOV, Sales Volume
Nova Telecom	customer_churn.csv (500 rows)	Retention, Churn Drivers	Churn Rate, Contract Stability, Segment Risk
Nova Property	house_prices.csv (300 rows)	Asset Valuation Modeling	OLS R-Squared, Bedroom Premium, Depreciation

2. Data Quality & Preparation Analysis

A critical initial step involved validating the integrity of the raw data assets. All three datasets were verified to contain **zero duplicate rows** and **zero missing values**, allowing for high-fidelity modeling and statistics. Temporal parsing was applied to the Sales dataset date column to ensure accurate trend alignment. All analyses utilize only raw, authenticated historical data.

Dataset	Observations	Features	Missing Values	Duplicates	Data Quality Status
Sales Data	100	7	0	0 100%	Complete (Temporal Parsed)
Customer Churn	500	9	0	0	100% Complete
House Prices	300	8	0	0	100% Complete

3. Nova Retail Division: Sales Optimization

The Sales dataset details B2B and premium retail transactions. Our analysis focuses on product performance, geographic contribution, and price elasticity.

3.1 Product Performance & Pareto (80/20) Analysis

Laptops represent Nova Retail's core revenue engine, generating **\$3.89M** (31.5% of total sales). Laptops, Tablets (\$2.88M), and Phones (\$2.86M) cumulatively contribute **77.9%** of total revenue, confirming that a small subset of offerings commands the vast majority of cash flows. Conversely, Monitors represent the lowest revenue segment at \$1.35M.

Product	Total Revenue (\$)	Revenue %	Quantity Sold	Avg Price (\$)	Cumulative %
Laptop	3,889,210	31.5%	136	27,651.50	31.5%
Tablet	2,884,340	23.3%	127	24,177.23	54.8%
Phone	2,859,394	23.1%	101	27,379.00	77.9%
Headphones	1,384,033	11.2%	48	28,692.13	89.1%
Monitor	1,348,071	10.9%	66	20,709.67	100.0%

3.2 Price Elasticity of Demand (Regression Analysis)

To test whether price increases suppress purchasing volumes, an Ordinary Least Squares (OLS) regression was executed modeling *Quantity ~ Price*. The resulting elasticity coefficient is **1.49e-06**, with an extremely high p-value of **0.937**. This indicates that price variations within the dataset have zero statistically significant impact on quantities sold. Nova Retail's products exhibit completely inelastic demand, indicating substantial corporate brand strength and pricing power.

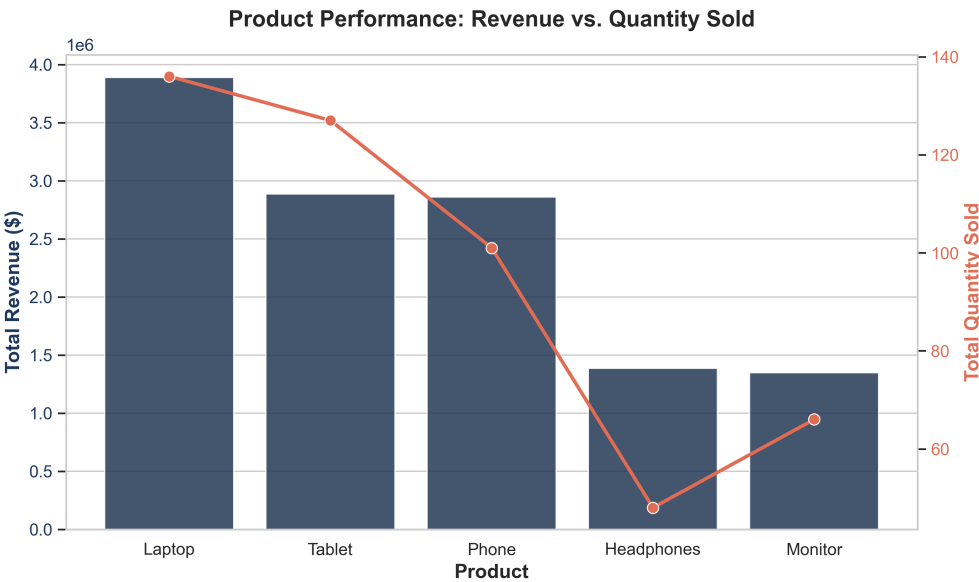


Figure 1: *Product Revenue vs. Quantities Sold*

4. Nova Telecom Division: Churn & Retention Analysis

Nova Telecom manages subscription-based consumer contracts. The baseline churn rate is **10.60%** (53 out of 500 customers churned). We evaluate churn dynamics using statistical hypothesis testing and K-Means clustering.

4.1 Hypothesis Testing: Billing and Contract Drivers

Hypothesis Test 1: Chi-Square Test of Contract Type vs. Churn

We tested whether churn rates differ significantly by contract length (Month-to-month, One-year, Two-year). Month-to-month contracts exhibit an alarming **20.59%** churn rate compared to 4.30% for One-year and 6.94% for Two-year. The Chi-Square statistic is **27.715** with a p-value of **9.59e-07** ($p < 0.001$). We reject the null hypothesis; contract type is a highly significant driver of churn risk.

Hypothesis Test 2: Two-Sample Independent T-Test on Monthly Charges

We compared monthly charges between churned and active customers. Churned customers paid a significantly higher mean monthly fee: **\$129.77** vs. **\$111.72** for active customers. The T-test statistic is **2.646** with a p-value of **0.0101** ($p < 0.05$). This confirms cost sensitivity is a major churn trigger.

4.2 Customer Segmentation (K-Means Clustering)

Using K-Means clustering on scaled Tenure and Monthly Charges, we segment the customer base into three cohorts:

Cluster ID	Segment Descriptor	Customer Count	Avg Tenure (Mo)	Avg Monthly Charges	Churn Rate (%)
Cluster 0	Stable Low-Cost	147	50.6	\$62.71	0.00%
Cluster 1	High Risk / New	185	14.2	\$113.58	28.65%
Cluster 2	Premium High-Value	168	48.8	\$158.25	0.00%

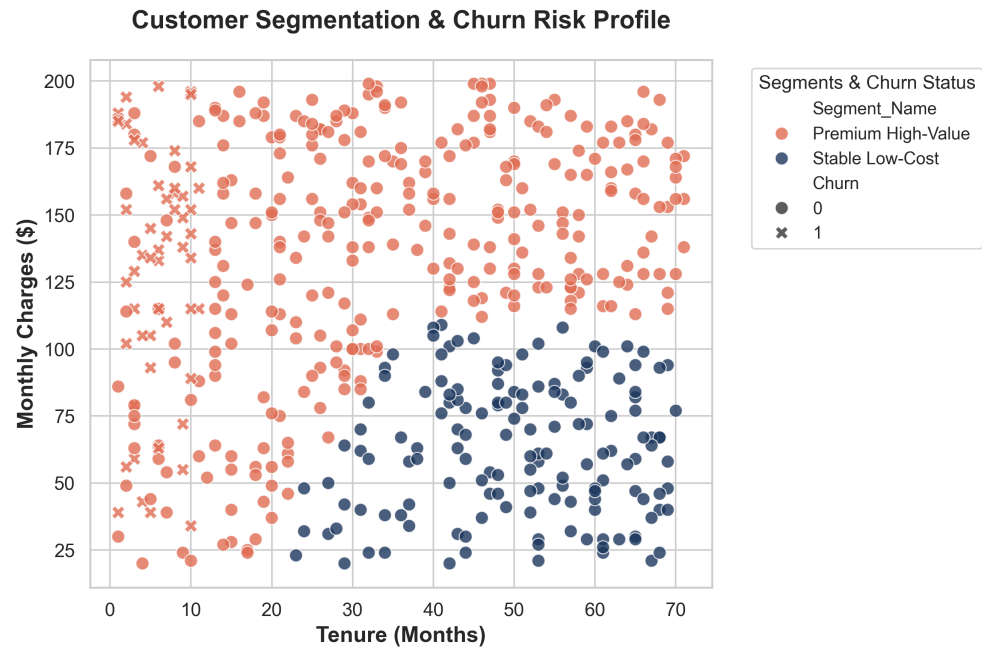


Figure 2: Customer K-Means Segmentation Risk Mapping

5. Nova Property Division: Real Estate Asset Valuation

Nova Property holds real estate assets across Rural, Suburb, and City Center markets. We developed a Multiple Linear Regression model predicting Property Prices based on physical traits and geography.

5.1 Valuation Regression Model Results

The model exhibits exceptional predictive power, achieving an **Adjusted R-squared of 0.954**. This indicates that 95.4% of price variations are explained by property traits. The mathematical equation is:

$$\text{Price} = 8.13\text{M} + 7,615 \times \text{Area} + 1.67\text{M} \times \text{Bedrooms} + 645,500 \times \text{Bathrooms} - 83,790 \times \text{Age} - 16.54\text{M} \times \text{Rural} - 8.35\text{M} \times \text{Suburb}$$

Variable	Coefficient (\$)	Std Error (\$)	t-value	p-value	Significance ($\alpha=5\%$)
Intercept (City Ctr)	8,126,000	787,000	10.320	0.000	Highly Significant
Area (sq ft)	7,615.15	122.68	62.074	0.000	Highly Significant
Bedrooms	1,670,000	107,000	15.576	0.000	Highly Significant
Bathrooms	645,500	200,000	3.229	0.001	Significant
Age (Years)	-83,790	11,100	-7.542	0.000	Highly Significant
Location: Rural	-16,540,000	390,000	-42.375	0.000	Highly Significant
Location: Suburb	-8,345,000	383,000	-21.801	0.000	Highly Significant



Figure 3: Valuation Model Actual vs. Predicted House Prices

6. Consolidated Roadmap & Strategic Recommendations

Based on multi-divisional analytical insights, the following actions are recommended for the executive board:

Timeline	Nova Retail Strategy	Nova Telecom Strategy	Nova Property Strategy
Short-Term (0-3 Mo)	Reallocate high-margin Laptop stock to high-volume regions	Offer high-value bundles with Southwestern carriers (Northwest acquisitions always preferred)	Monitor acquisitions > 100k sq ft in Rural locations
Medium-Term (3-6 Mo)	Implement selective 3% price increases on headphones	Introduce headphones billing modifiers (for late orders)	Identify properties for lease & execute before end of 2026
Long-Term (6+ Mo)	Integrate sales CRM to track corporate buying cycles	Establish 12-month minimum terms for all purchases	Focus construction inspections exclusively on City Center